Course Title: Cryptography and Network Security
Code: PCCCS601 & PEC-CS801B

Module 1: Syllabus: Attacks on Computers & Computer Security - Introduction, Need for Security, Security approaches, Principles of Security, Types of attack

Introduction:

This is the age of universal electronic connectivity, where the activities like hacking, viruses, electronic fraud are very common. Unless security measures are taken, a network conversation or a distributed application can be compromised easily.

Some simple examples are:

- Online purchases using a credit/debit card.
- A customer unknowingly being directed to a false website.
- A hacker sending a message to a person pretending to be someone else.

Network Security has been affected by two major developments over the last several decades. First one is introduction of computers into organizations and the second one being introduction of distributed systems and the use of networks and communication facilities for carrying data between users & computers. These two developments lead to 'computer security' and 'network security', where the computer security deals with collection of tools designed to protect data and to thwart hackers. Network security measures are needed to protect data during transmission. But keep in mind that, it is the information and our ability to access that information that we are really trying to protect and not the computers and networks.

Why We Need Information Security?

Because there are threats

Threats

A threat is an object, person, or other entity that represents a constant danger to an asset

The 2007 CSI survey

- 494 computer security practitioners
- 46% suffered security incidents
- 29% reported to law enforcement
- Average annual loss $350,424
- 1/5 suffered _targeted attack'
- The source of the greatest financial losses?
- Most prevalent security problem
- Insider abuse of network access
- Email

Threat Categories

- Acts of human error or failure
- Compromises to intellectual property

- Deliberate acts of espionage ortrespass
- Deliberate acts of informationextortion
- Deliberate acts of sabotage or vandalism
- Deliberate acts of theft
- Deliberate software attack
- Forces of nature
- Deviations in quality of service
- Technical hardware failures or errors
- Technical software failures or errors
- Technological obsolesce

Definitions

- **Computer Security** - generic name for the collection of tools designed to protect data and to thwart hackers
- **Network Security** - measures to protect data during their transmission
- **Internet Security** - measures to protect data during their transmission over a collection of interconnected networks
- our focus is on **Internet Security**
- which consists of measures to deter, prevent, detect, and correct security violations that involve the transmission & storage of information

Aspects of Security

- consider 3 aspects of information security:
  - **Security Attack**
  - **Security Mechanism**
  - **Security Service**

Security Attack

- any action that compromises the security of information owned by an organization
- information security is about how to prevent attacks, or failing that, to detect attacks on information-based systems
- often *threat* & *attack* used to mean same thing
- have a wide range of attacks

- can focus of generic types of attacks
  - Passive
  - Active

INTERRUPTION

An asset of the system is destroyed or becomes unavailable or unusable. It is an attack on availability.

Examples:

- Destruction of some hardware
- Jamming wireless signals
- Disabling file management systems

INTERCEPTION

An unauthorized party gains access to an asset. Attack on confidentiality.

Examples:

- Wire tapping to capture data in a network.
- Illicitly copying data or programs
- Eavesdropping

MODIFICATION

When an unauthorized party gains access and tampers an asset. Attack is on Integrity.

Examples:

- Changing data file
- Altering a program and the contents of a message

FABRICATION

An unauthorized party inserts a counterfeit object into the system. Attack on Authenticity. Also called impersonation

Examples:

- Hackers gaining access to a personal email and sending message
- Insertion of records in data files
- Insertion of spurious messages in a network

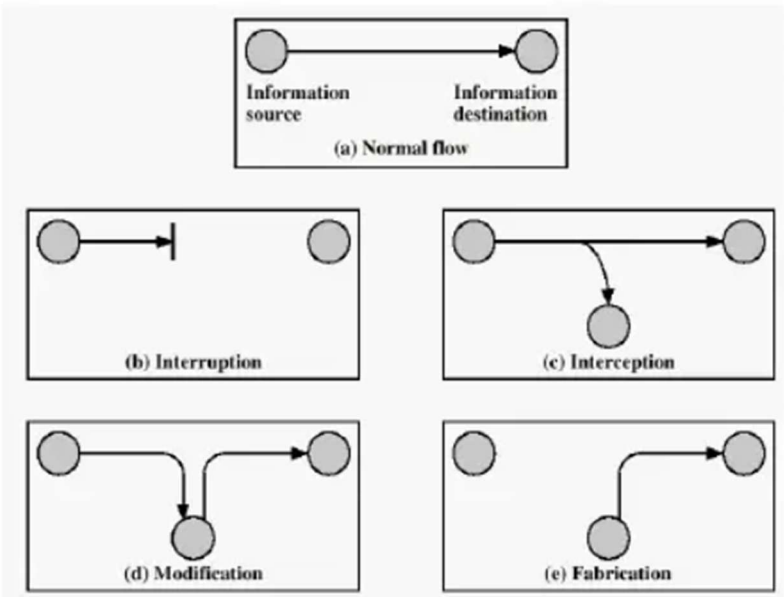

Figure 1.1 Security Threats

SECURITY SERVICES

It is a processing or communication service that is provided by a system to give a specific kind of production to system resources. Security services implement security policies and are implemented by security mechanisms.

Confidentiality

Confidentiality is the protection of transmitted data from passive attacks. It is used to prevent the disclosure of information to unauthorized individuals or systems. It has been defined as "ensuring that information is accessible only to those authorized to have access".The other aspect of confidentiality is the protection of traffic flow from analysis.
Ex: A credit card number has to be secured during online transaction.

Authentication

This service assures that a communication is authentic. For a single message transmission, its function is to assure the recipient that the message is from intended source. For an ongoing interaction two aspects are involved. First, during connection initiation the service assures the authenticity of both parties. Second, the connection between the two hosts is not interfered allowing a third party to masquerade as one of the two parties. Two specific authentication services defines in X.800 are

Peer entity authentication: Verifies the identities of the peer entities involved in communication. Provides use at time of connection establishment and during data transmission. Provides confidence against a masquerade or a replay attack
Data origin authentication: Assumes the authenticity of source of data unit, but does not provide protection against duplication or modification of data units. Supports applications like electronic mail, where no prior interactions take place between communicating entities.

Integrity

Integrity means that data cannot be modified without authorization. Like confidentiality, it can be applied to a stream of messages, a single message or selected fields within a message. Two types of integrity services are available. They are

Connection-Oriented Integrity Service: This service deals with a stream of messages, assures that messages are received as sent, with no duplication, insertion, modification, reordering or replays. Destruction of data is also covered here. Hence, it attends to both message stream modification and denial of service.

Connectionless-Oriented Integrity Service: It deals with individual messages regardless of larger context, providing protection against message modification only.

An integrity service can be applied with or without recovery. Because it is related to active attacks, major concern will be detection rather than prevention. If a violation is
detected and the service reports it, either human intervention or automated recovery machines are required to recover.

Non-repudiation

Non-repudiation prevents either sender or receiver from denying a transmitted message. This capability is crucial to e-commerce. Without it an individual or entity can deny that he, she or it is responsible for a transaction, therefore not financially liable.

Access Control

This refers to the ability to control the level of access that individuals or entities have to a network or system and how much information they can receive. It is the ability to limit and control the access to host systems and applications via communication links. For this, each entity trying to gain access must first be identified or authenticated, so that access rights can be tailored to theindividuals.

Availability

It is defined to be the property of a system or a system resource being accessible and usable upon demand by an authorized system entity. The availability can significantly be affected by a variety of attacks, some amenable to automated counter measures i.e authentication and encryption and others need some sort of physical action to prevent or recover from loss of availability of elements of a distributed system.